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3 Pollen grains germinate in a solution that contains sucrose and boric acid.

The effect of boric acid concentration on the percentage germination of pollen grains was studied.

(a) State the dependent variable in this study.

(1)

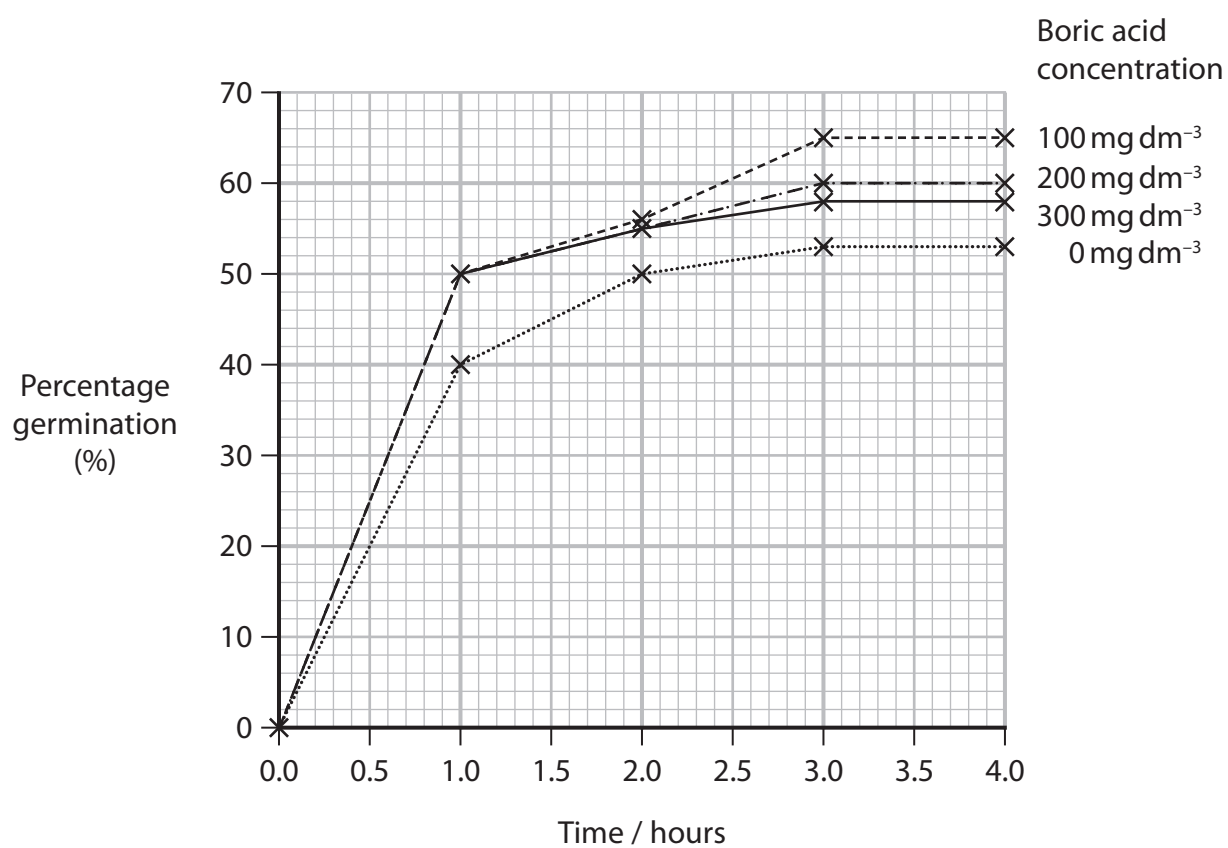
(b) In this study, the solutions used were maintained at pH 6.

Explain why this pH was maintained.

(2)



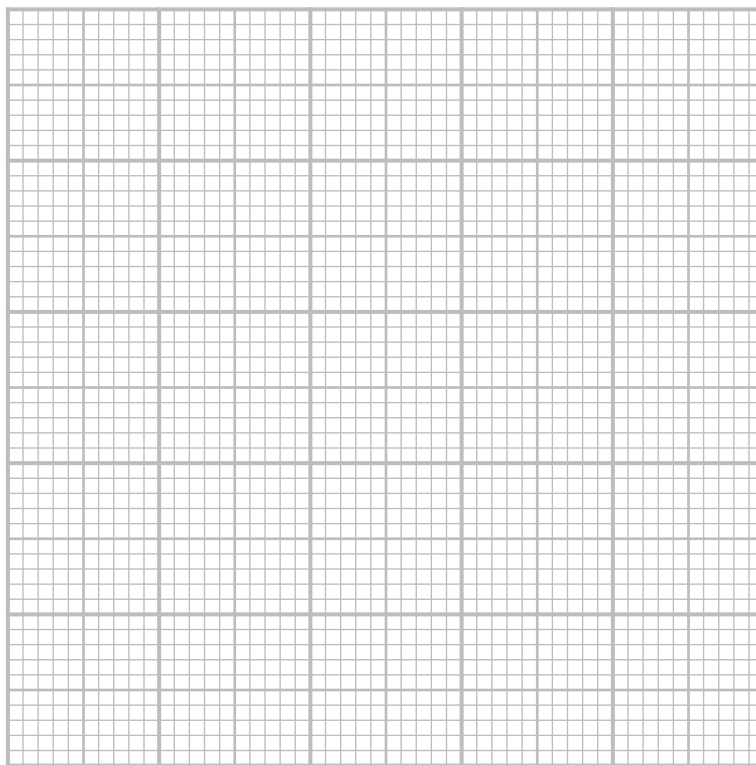
(c) The graph shows the results of this study.



- (i) Draw a graph to show how boric acid concentration affects germination at 4 hours.

Join the points with straight lines.

(5)



- (ii) Describe the effect of boric acid concentration on pollen grain germination at 4 hours.

(2)

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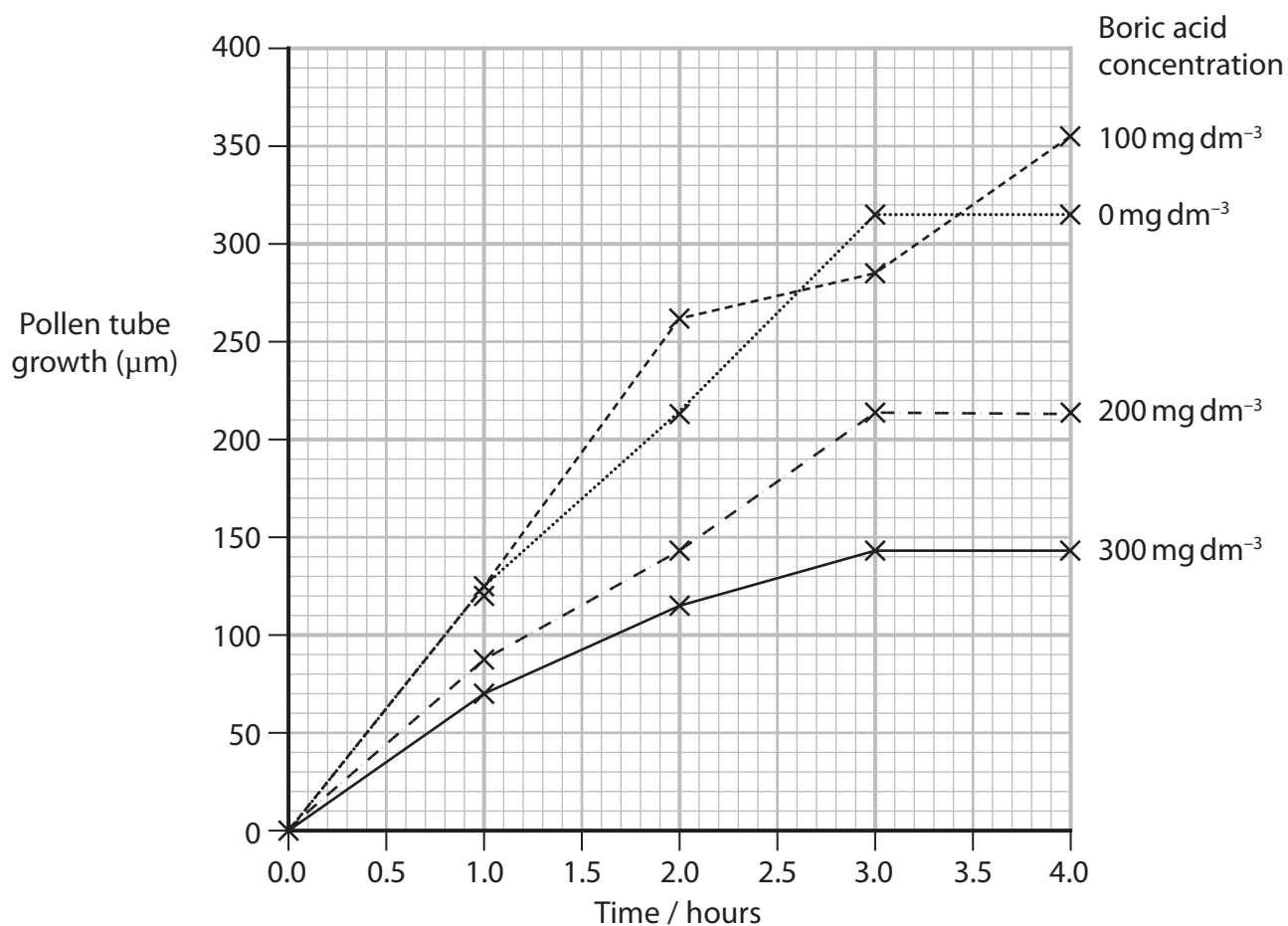
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- (d) In another study, the effect of boric acid concentration on pollen tube growth was investigated.

The graph shows the results of this study.



- (i) During the first two hours, the rate of pollen tube growth at a boric acid concentration of 100 mg dm^{-3} is $130.5 \mu\text{m hour}^{-1}$.

Calculate the percentage increase in the rate of pollen tube growth between the rate at 0 mg dm^{-3} and the rate at 100 mg dm^{-3} , during the first two hours.

(2)

..... %



(ii) Describe how the results shown in the graph could have been obtained.

You are provided with a 500 mg dm^{-3} boric acid solution.

(4)

(e) Compare and contrast the effect of boric acid concentration on pollen grain germination with its effect on pollen tube growth.

Use the information given in this question.

(2)

(Total for Question 3 = 18 marks)

TOTAL FOR PAPER = 50 MARKS

